Ex: The cetane number is a critical property in specifying the ignition quality of a fuel used in a disesel engine. The article "Relating the cetane Number of Biodiesel Fuels to Their Faulty Acid Composition: A Critical Study" (J. of Automobile Eng., 2009; 565-583) included the following data on x = iodine value (g) and y = cetane number for a sample of 14 biofuels.

The iodine value is the amount of iodine necessary to saturate a sample of 100g of oil.

x:	132.0	129.0	120.0	113.2	105.0	92.0	
y:	46.0	48.0	51.0	52.1	54.0	52.0	
x:	84.0	83.2	88.4	59.0	80.0	81.5	71.0
y:	59.0	58.7	61.6	64.0	61.4	54.6	58.8
x:	69.2						
y:	58.0						

Fit a straight line of the form $y = \beta_0 + \beta_1 x$ to these values by the method of least square. Estimate the cetane number when the iodine value is 100.

Soln: $\Sigma x_i = 1307.5$, $\Sigma y_i = 779.2$

$\Sigma x_i^2 = 128,913.93$, $\Sigma y_i^2 = 43,745.22$

$\Sigma x_i y_i = 71,347.30$

$$\bar{x} = \frac{1307.5}{14} = 93.3928, \quad \bar{y} = \frac{779.2}{14} = 55.6571$$

$$S_{xx} = \Sigma(x_i - \bar{x})^2 = \Sigma x_i^2 - n\bar{x}^2 = 6802.7693$$

$$S_{yy} = \Sigma(y_i - \bar{y})^2 = \Sigma y_i^2 - n\bar{y}^2 = -1424.4143$$

$$\hat{\beta}_1 = \frac{\Sigma x_i y_i - n\bar{x}\bar{y}}{\Sigma x_i^2 - n\bar{x}^2} = \frac{S_{xy}}{S_{xx}} = \frac{-1424.4143}{6802.7693}$$

$$= -0.2094$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$= 55.6571 - (-0.2094)(93.3928)$$

$$= 75.2124$$

$$y = \hat{\beta}_0 + \hat{\beta}_1 x$$

or, $y = 75.2124 - 0.2094x$

2nd part: A prediction of the cetane number when the iodine value is 100 is

$$\hat{y} = 75.212 - 0.2094(100) = 54.27$$

Coefficient of Multiple Correlation

* In a tri-variate distribution in which each of the variables x_1, x_2 and x_3 has n observations, the multiple correlation coefficient of x_1 on x_2 and x_3 usually denoted by $R_{1\cdot 23}$ is the simple correlation coefficient between x_1 and the joint effect of x_2 and x_3 on x_1. In other words, $R_{1\cdot 23}$ is the correlation coefficient between x_1 and its estimated value as given by the plane of regression of x_1 on x_2 and x_3

$$R^2_{1\cdot 23} = \frac{r^2_{12} + r^2_{13} - 2 r_{12} r_{13} r_{23}}{1 - r^2_{23}}$$

$$0 \leq R_{1\cdot 23} \leq 1$$

- **Coefficient of Partial correlation:**

Sometimes the correlation between two variables x_1 and x_2 may be partly due to the correlation of a third variable x_3 with both x_1 and x_2. In such a situation, one may want to know what the correlation between x_1 and x_2 would be if the effect of x_3 on each of x_1 and x_2 were eliminated.

~~Thi~~ - This correlation is called the partial correlatio

$$\boxed{r_{12\cdot 3} = \frac{r_{12} - r_{13}\, r_{23}}{\sqrt{(1-r_{13}^2)(1-r_{23}^2)}}}$$

r_{12}

$\underline{r_{12\cdot 3}}$